0.1 title: “EcoSphere AI” subtitle: “AI-Powered Sustainability Mission Control” author: “Crimson Syndicate — AIEM IDEAS 2026”

Registration and project dossier. This document is designed as a form-ready, competition-ready summary for AIEM IDEAS 2026. It describes EcoSphere AI as a **working campus-pilot prototype**. Simulated data, modelled outputs and unconfigured external services are explicitly distinguished from live, verified operations.

0.2 Project Snapshot

Item	Details
Proposed Team Name	**Crimson Syndicate** — please confirm this exact name before submitting the registration form.
Idea Title	**EcoSphere AI — AI-Powered Sustainability Mission Control**
Idea Category	Artificial Intelligence, Sustainability and Smart Campus Technology
Primary SDG	SDG 13 — Climate Action
Supporting SDGs	SDG 7 — Affordable and Clean Energy; SDG 9 — Industry, Innovation and Infrastructure; SDG 11 — Sustainable Cities and Communities; SDG 12 — Responsible Consumption and Production
Primary Beneficiaries	Campus administrators, facilities teams, maintenance teams, sustainability clubs and students
Prototype Positioning	A traceable sustainability-operations prototype for an AIEM Campus pilot, with controlled simulated demonstration data when live metering is not available.

0.3 Problem Statement

Educational campuses and institutions often receive energy, water and waste information through disconnected spreadsheets, manual records or delayed utility bills. This fragmented approach makes unusual resource consumption difficult to identify early, limits accountability for sustainability actions and makes it hard for decision-makers to compare which intervention will create the highest environmental impact for the available budget.

The result is reactive sustainability management: a potentially avoidable HVAC spike, water leak or waste increase may be noticed only after it has already increased operational cost and carbon emissions.

Institutions need a practical system that connects measurement, anomaly awareness, transparent calculations, intervention planning and evidence-led action in one understandable workflow.

0.4 Proposed Solution

EcoSphere AI is an AI-powered sustainability mission-control platform designed for a campus-pilot setting. It brings energy, water and waste readings into one governed operational workflow and turns them into traceable indicators, anomaly evidence, carbon calculations, EcoScore snapshots, forecasts and clearly labelled sustainability scenarios.

The product follows a continuous operating loop:

Monitor → Detect → Predict → Simulate → Recommend → Act → Measure → Repeat

Its numerical calculations are deterministic and server-owned. AI is used to explain evidence and organise recommendations; it does not invent consumption, savings, carbon or score values. This design supports more credible sustainability discussions, because every recommendation is connected to recorded or explicitly simulated evidence.

0.5 Core Capabilities

Capability	How EcoSphere AI supports the campus pilot
Sustainability monitoring	Organises energy, water and waste readings against tenant, site and meter records.
Data-quality controls	Validates imports and readings, quarantines problematic data and preserves source lineage.
Carbon and EcoScore evidence	Applies deterministic calculation logic to create transparent carbon and EcoScore snapshots.
Anomaly detection	Detects unusual consumption patterns and creates evidence-linked alerts.
Forecasting and trends	Produces short-horizon, clearly limited forecasts from available evidence.
What-if simulator	Models energy, renewable-energy, water, waste, recycling and investment choices without claiming realised outcomes.
Intervention comparison	Compares actions such as LED upgrades, smart HVAC controls, solar installation and water-saving systems using modelled impact, cost, savings and SDG contribution.
Action and evidence workflow	Records accountable sustainability actions, approvals and evidence status inside the tenant workspace.

Capability	How EcoSphere AI supports the campus pilot
Controlled live demonstration	Uses explicitly labelled simulated pilot data to demonstrate monitoring cycles and an HVAC-spike anomaly safely.

0.6 Innovation and Differentiation

EcoSphere AI is not only a dashboard. It is designed as a **decision-support and evidence workflow** for sustainability operations. Its difference lies in connecting three layers that are commonly separated: trusted operational data, deterministic environmental calculations and user-friendly intervention planning.

First, its monitoring architecture is browser-independent: the background monitoring worker is designed to process eligible readings without requiring the dashboard browser to remain open. Second, the platform distinguishes live evidence from simulated demo data and modelled scenarios. Third, its AI layer is bounded to explanation and recommendation reasoning, while calculations remain reproducible and traceable. This makes the concept suitable for an initial campus pilot and for future expansion to hostels, laboratories, offices and community infrastructure.

0.7 Expected Beneficiaries and Value

Campus facility and maintenance teams can use EcoSphere AI to identify potentially unusual resource consumption earlier and document follow-up actions. Administrators can use one executive view to examine resource trends, active alerts, EcoScore evidence and sustainability priorities. Students and sustainability clubs can use the platform to support transparent climate-action discussions, campaigns and pilot projects.

The approach is also relevant to hostels, educational institutions, offices and local public facilities that need a structured, affordable route from disconnected utility data to evidence-led resource-management decisions.

0.8 Sustainable Development Goal Alignment

SDG	EcoSphere AI contribution
SDG 13 — Climate Action	Supports lower-emission operational decisions through carbon evidence, anomaly awareness and modelled intervention analysis.
SDG 7 — Affordable and Clean Energy	Tracks electricity-related evidence and compares energy-efficiency and renewable-energy interventions.
SDG 9 — Industry, Innovation and Infrastructure	Demonstrates a data-driven, traceable monitoring architecture for institution-scale infrastructure decisions.
SDG 11 — Sustainable Cities and Communities	Provides a campus model that can be adapted to institutional and community infrastructure.

SDG	EcoSphere AI contribution
SDG 12 — Responsible Consumption and Production	Connects water, waste and resource-consumption evidence with accountable improvement actions.

0.9 AIEM Campus Demonstration Flow

The ten-minute demonstration is structured to make the use case understandable quickly and honestly:

1. Open the AIEM Campus dashboard and establish the EcoScore, monitored meters and sustainability indicators.
2. Start the clearly labelled live simulation so controlled pilot readings progress through the monitoring pipeline.
3. Inject a controlled HVAC energy spike.
4. Show the anomaly, alert, EcoScore response and evidence-linked AI explanation.
5. Open the What-If Sustainability Simulator and model a 15% energy reduction.
6. Compare projected carbon reduction, estimated savings and intervention trade-offs.
7. Present the highest-impact intervention recommendation and its SDG alignment.
8. Close with the roadmap from controlled simulation to an approved, live AIEM Campus meter-data pilot.

0.10 Implementation Approach and Responsible Boundaries

EcoSphere AI includes a React-based user interface, an Express/tRPC backend, tenant-scoped data handling, a MySQL/TiDB-compatible persistence model and a browser-independent monitoring worker design. CSV import, controlled simulation, alerts, scenario comparison, reports, action tracking and operational administration are represented as integrated product workflows.

For accuracy, the team should not claim that the project currently has live AIEM telemetry, certified emissions accounting, realised savings, a live Odoo integration, autonomous external notification delivery or a production scheduler unless those capabilities are separately configured, deployed and verified. Demonstration readings and outcomes are labelled as simulated or modelled where applicable.

0.11 Team Details

Role	Name	Roll Number	Class	Department
Project Leader	Shivam Gawade	24ec25	TE	Computer Engineering
Member 2	Rahul Ravi Rathod	24ec17	TE	Computer Engineering
Member 3	Rehan Harmalkar	24ec18	TE	Computer Engineering
Member 4	Ashwith Shetty	24co35	TE	Computer Engineering

Role	Name	Roll Number	Class	Department
Member 5	Deekshith TS	23ec12	BE	Computer Engineering

Form item still required: enter Shivam Gawade's active WhatsApp number in the Project Leader contact field before submission. Verify spelling, roll numbers, class and department format against the college record before submitting.

0.12 Form-Ready Responses

Form field	Paste this response
Team Name	Crimson Syndicate *(confirm with the team before submission)*
Title of the Idea	EcoSphere AI — AI-Powered Sustainability Mission Control
Idea Category	Artificial Intelligence, Sustainability and Smart Campus Technology
Problem Statement	Campuses and institutions often record energy, water and waste information in disconnected spreadsheets or only after a bill is received. This makes abnormal consumption difficult to identify early, limits accountability for sustainability actions, and makes it hard for decision-makers to compare the likely impact of interventions such as efficient lighting, smart HVAC controls or solar adoption. EcoSphere AI addresses this gap through a single sustainability operations platform that converts structured meter readings into traceable environmental metrics, anomaly alerts, EcoScore evidence, short-horizon forecasts and clearly labelled what-if scenarios. It helps campus teams move from delayed reporting to evidence-led monitoring, prioritisation and action.
Which person/sector will your idea benefit?	Educational institutions and campus facility teams are the primary beneficiaries, especially administrators, maintenance teams, sustainability clubs and students who need a shared view of resource consumption. The approach can also benefit hostels, offices,

Form field	Paste this response
	municipal facilities and small institutions that need an affordable way to monitor energy, water, waste and carbon evidence, identify unusual consumption, compare sustainability interventions and communicate progress responsibly.
SDGs to select	Affordable and Clean Energy; Industry Innovation and Infrastructure; Sustainable Cities and Communities; Responsible Consumption and Production; Climate Action

0.13 Registration Resources

Resource	Link and intended use
Participants WhatsApp group	Join using the supplied participant link. The team should join only through the official link supplied by the organisers.
Official PPT template	Open the supplied Google Slides template. Download a personal copy; do not edit or share the original template.

0.14 Pre-Submission Checklist

1. Confirm the team name, team roster, roll numbers and department labels.
2. Add the Project Leader's active WhatsApp number.
3. Select only the five SDGs listed in this dossier.
4. Use the form-ready project text above, retaining the "pilot prototype" framing.
5. Check the email account shown in the form before submission.
6. Submit only after every team member has verified their personal details.